

AVINASH RAJ

B.Tech. Computer Science & Engineering | SISTec GN, Bhopal

+91-7295053251 • aviraj22072005@gmail.com • Bhopal, Madhya Pradesh, India

LinkedIn: [linkedin.com/in/avinash-raj-30797b300](https://www.linkedin.com/in/avinash-raj-30797b300) • GitHub: github.com/Avinash58 • Portfolio: avinash58.github.io/MyPortfolio

PROFESSIONAL SUMMARY

Computer Science & Engineering undergraduate with a strong foundation in Data Structures & Algorithms, Object-Oriented Programming, and Python-based software development. Hands-on experience building REST APIs, full-stack web applications, and database-driven systems using Python, Flask, FastAPI, SQL, JavaScript, TypeScript, and React/Next.js, including client-server architecture, MVC design, real-time dashboards, and LLM-powered (Google Gemini) features. Strong problem-solving and collaboration skills developed through team projects, hackathons, and certifications; a quick learner eager to apply these skills to real-world software engineering problems.

TECHNICAL SKILLS

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Client-Server Architecture, Distributed Systems (academic/project-level), Agile Methodology

Languages: Python, SQL, JavaScript, TypeScript, HTML5, CSS3

Frameworks & Libraries: FastAPI, Flask, REST APIs, Next.js, React, Tailwind CSS, SQLAlchemy, Pydantic

AI & APIs: Google Gemini API (structured output), Google Places / Geocoding / Maps APIs, JWT Authentication

Databases: PostgreSQL, SQLite, MySQL, Redis (DBMS concepts)

Tools & Platforms: Git, GitHub, Docker / Docker Compose, Nginx, VS Code, Postman, Linux (basic)

Networking: Cisco Networking Academy Certified, HTTP request-response / client-server communication

Other: Salesforce (basic), Quantum Computing (awareness), Data Analytics

PROJECTS

AI-Powered City Search & Discovery Platform

2026

Tech Stack: Python, FastAPI, Next.js, React, TypeScript, PostgreSQL, Redis, Docker, Gemini API | *GitHub:* github.com/Avinash58

- Built a full-stack platform where users search restaurants, hotels and attractions in any city, using **Google Gemini 2.5 Flash** with schema-validated JSON output and a multi-tier fallback (Gemini → OpenRouter → curated data), plus a multi-turn AI travel chatbot.
- Integrated Google Places & Geocoding APIs with Haversine-based proximity ranking and browser geolocation for "near me" search.
- Developed a versioned REST API (JWT + bcrypt auth, role-based admin panel, rate limiting) and containerized 6 services (PostgreSQL, Redis, Elasticsearch, FastAPI, Next.js, Nginx) with Docker Compose.

CivicConnect — Civic Issue Reporting System

Oct 2025 - Jan 2026

Tech Stack: Python, Flask, SQLite, HTML5, CSS3, JavaScript, REST API | *Team:* 4 | *Mentor:* Dr. Amit Dubey | *GitHub:* github.com/Avinash58

- Architected a scalable client-server platform with REST APIs and a modular MVC backend, enabling citizens to report and track urban issues (potholes, drainage, streetlights) in place of manual complaint processes.
- Built a real-time admin dashboard that let authorities prioritize and update issue statuses, cutting average complaint filing time versus the prior paper-based workflow.

EDUCATION

Sagar Group of Institutions — SISTec GN, Bhopal

2023–2027

B.Tech. — Computer Science & Engineering, CGPA: 7.1 / 10

Modern Academy Ashok Nagar, Gaya — 12th, CBSE

2023

D.A.V C.R.R.C. Medical, Gaya — 10th, CBSE

2021

CERTIFICATIONS & ASSESSMENTS

- Python Essentials 1 & 2 — Cisco Networking Academy: core Python programming, data structures, OOP, and scripting.
- Data Analytics Job Simulation (Deloitte) and Quantum Computing (C-DAC & IIT Roorkee) — data interpretation, business analytics, and foundational quantum algorithms.

HACKATHONS, SEMINARS & WORKSHOPS

- Samadhan 2.0 — National Level Hackathon (Oct 15–17, 2025), SISTec GN.
- AICTE-IDE National Bootcamp 2026 (Apr 20–24, 2026), MITS — innovation & design engineering bootcamp.
- Smart India Hackathon (SIH) 2024 — national-level government-sponsored hackathon.
- 7th International Conference on Image Processing & Pattern Recognition (Dec 19–20, 2024), MANIT, Bhopal.

CO-CURRICULAR & EXTRA-CURRICULAR ACTIVITIES

- Attended AI/ML workshops, contributed to team-based technical presentations, and engaged in automation experiments and hands-on AI prototyping outside the curriculum.

Languages: English, Hindi | *Address:* SISTEC GN Campus, Bhopal, MP — 462036